

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO3: *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

Assessment Tool Weightage: 10%

**Duration :60 Mins
On Class**

QUESTIONS: 50

Total Marks:50

Annexure A

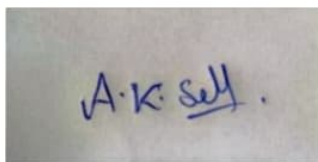
SCENARIO BASED TASK

Code of Conduct:

I A K Selva Prabhu(192421224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the handwritten signature "A.K. Selva" in blue ink.

1. Desk Calculator using SD

Grammar :

$$S \rightarrow En$$

$$E \rightarrow E + T \mid E - T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid \text{digit}$$

Input: $(6+4)*(4+3)$

$$\text{SD} : E \rightarrow E_1 + T \{ E.\text{val} = E_1.\text{val} + T.\text{val} \}$$

$$E \rightarrow E_1 - T \{ E.\text{val} = E_1.\text{val} - T.\text{val} \}$$

$$E \rightarrow T \{ E.\text{val} = T.\text{val} \}$$

$$T \rightarrow T_1 * F \{ T.\text{val} = T_1.\text{val} * F.\text{val} \}$$

$$T \rightarrow T_1 / F \{ T.\text{val} = T_1.\text{val} / F.\text{val} \}$$

$$T \rightarrow F \{ T.\text{val} = F.\text{val} \}$$

$$F \rightarrow (E) \{ F.\text{val} = E.\text{val} \}$$

$$F \rightarrow \text{digit} \{ F.\text{val} = \text{digit}.\text{val} \}$$

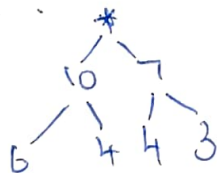
Attribute Evaluation

$$6 + 4 = 10$$

$$4 + 3 = 7$$

$$10 * 7 = 70$$

Annotated tree :

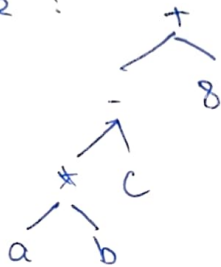
 $\therefore$ Final value = 70

2. Expression: $a * b - c + 8$

* has higher precedence than - and +

$$\therefore (a * b) - c + 8$$

Syntax tree:



SDD

$$E \rightarrow E_1 + T \quad \{ E.val = E_1.val + T.val \}$$

$$E \rightarrow E_1 - T \quad \{ E.val = E_1.val - T.val \}$$

$$E \rightarrow T \quad \{ E.val = T.val \}$$

$$T \rightarrow T_1 * F \quad \{ T.val = T_1.val * F.val \}$$

$$T \rightarrow F \quad \{ T.val = F.val \}$$

$$F \rightarrow id \quad \{ F.val = id.val \}$$

Evaluation sequence.

$$\begin{array}{c} a * b \\ \downarrow \\ (a * b) - c \\ \downarrow \\ ((a * b) - c) + 8 \end{array}$$

So, the correct interpretation is:

$$(a * b) - c + 8$$

3. L-Attributed Definition.

Given:

$$A \rightarrow PA$$

$$A \rightarrow XY$$

Each symbol has synthesized s and inherited i attributes

Rule 1

$$P.i = A.i + 2$$

$$A.i = P.i + A_i$$

$$A.s = P.s + A.s$$

Rule 1 is L-attributed.

Reason:

* $P.i$ depends on parent $A.i$.

Evaluation from left to right

$$A.i \rightarrow P.i \rightarrow P.s \rightarrow A.i \rightarrow A.s \rightarrow A.s$$

Rule 2

$$X.i = A.i + Y.s$$

$$Y.i = X.s + A.i$$

Rule 2 is not L-attributed.

Reason:

$$X.i = A.i + Y.s$$

Here $X.i$ depends on $Y.s$, but Y is to the right of X .

It violates the left to right condition

Rule 1 $\rightarrow$ L-attributed

Rule 2 $\rightarrow$ Not L-attributed.

S-Attributed SD and L-R Parsing

Grammar

$$S \rightarrow EN$$

$$E \rightarrow E + T \mid E - T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid \text{digit}$$

$$N \rightarrow ;$$

Input $2 + 5 * 6 ;$

Important Semantic Rules

$$E \rightarrow E_1 + T \{ E.val = E_1.val + T.val \}$$

$$E \rightarrow E_1 - T \{ E.val = E_1.val - T.val \}$$

$$E \rightarrow T \{ E.val = T.val \}$$

$$T \rightarrow T_1 * F \{ T.val = T_1.val * F.val \}$$

$$T \rightarrow T_1 / F \{ T.val = T_1.val / F.val \}$$

$$T \rightarrow F \{ T.val = F.val \}$$

$$F \rightarrow (E) \{ F.val = E.val \}$$

$$F \rightarrow \text{digit} \{ F.val = \text{digit}.val \}$$

LR Parsing - Simplified Trace.

$2 \rightarrow \text{Shift}$

$* \rightarrow \text{Shift}$

$6 \rightarrow \text{Reduce}$

$6 \rightarrow \text{Shift}$

$T \rightarrow \text{Reduce}$

$F \rightarrow \text{Reduce}$

$E \rightarrow \text{Reduce}$

$T \rightarrow \text{Reduce}$

$+ \rightarrow \text{Shift}$

$E \rightarrow \text{Reduce}$

$5 \rightarrow \text{Shift}$

$; \rightarrow \text{Shift}$

$F \rightarrow \text{Reduce}$

$S \rightarrow \text{Reduce}$

$T \rightarrow \text{Reduce}$

Accept.

Generated Intermediate Code.

A.K. Schenckel

Since * has higher precedence:

$$t_1 = 5 * 6$$

$$t_2 = 2 + t_1$$

$$\therefore t_1 = 30$$

$$t_2 = 32$$

$$\text{Final result} = 32.$$

Top-Down Translation.

$$S \rightarrow E_n$$

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid \text{digit}$$

$$\text{Input: } 3 * 4 + 8n$$

* has higher precedence.

$$(3 * 4) + 8$$

SDD:

$$E \rightarrow E_1 + T \{ E.val = E_1.val + T.val \}$$

$$E \rightarrow T \{ E.val = T.val \}$$

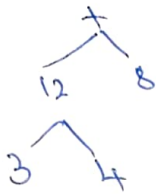
$$T \rightarrow T_1 * F \{ T.val = T_1.val * F.val \}$$

$$T \rightarrow F \{ T.val = F.val \}$$

$$F \rightarrow (E) \{ F.val = E.val \}$$

$$F \rightarrow \text{digit} \{ F.val = \text{digit.lex.val} \}$$

Attribute Evaluation



Attribute Evaluation

$$3 \rightarrow 3$$

$$4 \rightarrow 4$$

$$3 * 4 = 12$$

$$12 + 8 = 20$$

$$\therefore \text{S.val} = 20$$

$$\text{Final value} = 20.$$

~~Q1 $\Rightarrow$ 3~~

$$\text{Q2} \Rightarrow (3 * 4) + 8$$